Supplementary material

Classifier selection, score estimation and event coverage in inertial pavement screening

RoadSens fold means and pooled OOF scores are separate summaries. PVS values are individual-domain results or three-domain axis means; Mexico values concern two documented route families. Table identifiers are sequential within this supplement.

S1. Software, execution and provenance

The analysis environment is Python 3.13.12, scikit-learn 1.9.0, NumPy 2.5.2, pandas 3.0.5, joblib 1.5.3, SciPy 1.18.0 and threadpoolctl 3.6.0. Fits use at most four CPU threads. The execution matrix below records the original 1,580 canonical fit-folds; later extensions have separate inventories. Model reloads, source identities, split memberships, labels and numerical summaries are checked independently of the fitting drivers.

Source archives, exact split maps and model hashes are indexed in the execution records. The immutable RoadSens cache has SHA-256 `fa2078df1857f7872c13c6fe187f0883d85a23f9d15fdb3c6897a53314fdea16`. The RoadSens Figshare record uses citation year 2025 and reports version 3 publication on 24 February 2026. The analysis uses that version-specific payload (Khandakar et al. 2025b).

Table S1: Original execution matrix and authoritative output records.

Block	Fixed comparison	New canonical fit-folds	Authoritative output
A	Four models × four feature sets × three splits × two policies × five folds; seed 42	480	`../runs/main_A_v2/analysis/main_A_analysis.md`
R	RF seeds 43/44 × four features × random/recording × two policies × five folds	160	`../runs/consistent_runtime_R_v1/`; `../runs/auxiliary_summary_v1.md`
B	Three non-RF models × full/context × random/recording × two policies × seeds 43/44 × five folds	240	`../runs/phase_B_v1/`; `../runs/auxiliary_summary_v1.md`
C	RF × eight ablations × three splits × two policies × five folds	240	`../runs/phase_C_v2/independent_validation.md`
D	Three additional capacity/budget combinations × three splits × two policies × five folds	90	`../runs/phase_D_capacity_vs_A_v1.md`
E	Binary RF × four feature sets × three splits × five folds	60	`../runs/phase_E_v1/`; `../runs/auxiliary_summary_v1.md`
H	Specified component-heldout H1 (100) and byte-copy-removal H2 (90) fits	190	`../runs/content_overlap_v2/analysis/`
I	Two spatial grids × two models × three feature groups × five folds	60	`../runs/spatial_binary_v1/independent_summary.md`
S	STRIDE: two models × two feature sets × three seeds × five folds	60	`../runs/stride_supplement_v2/validation_summary.md`
Total	Includes120 A RF reproduction fits; excludes330 earlier exploratory fits	**1,580**	`../execution_inventory.json`

S2. Tasks and evaluation definitions

RoadSens is analysed as a three-class source-labelled task, with a separate explicit bump/pothole control. PVS supplies a binary speed-bump task, STRIDE a whole-record bump/pothole task, and Mexico/Puebla a released non-event/pothole/bump task. Each task retains its source annotation semantics and fitted models. Recording, content-component and spatial grouping represent distinct information-sharing restrictions. The nested PVS analysis compares candidate selection with estimation of the selected candidate's outer performance. These definitions guide interpretation of the tables below.

Availability-only predictors retain a mask only when its underlying numeric field has a finite fit-row value. In the concatenated 84-field numeric/mask representation, the finite-column rule is applied to the concatenated matrix and can retain an all-zero mask. Numeric imputation uses fit-row medians, including for contextual predictors; effective feature dimensions can therefore differ across folds. Exact field ordering follows the saved feature manifests.

S3. Same-runtime random-forest seed checks

The table reports full-numeric random-minus-recording fold-mean differences under seeds 42–44. Each configuration has five positive fold-index differences. Seed checks vary model randomness, split assignment and fit sampling within the same source cohort. The principal comparison uses seed 42.

Historical seed-43/44 RF models were serialised under scikit-learn 1.5.2. The reported same-runtime R block refits their saved source/row partitions under 1.9.0. The original assignments remain the split authority across software versions.

Table S2: Full-numeric random-minus-recording differences of five-fold mean macro-F1.

Policy	Seed 42	Seed 43	Seed 44
Primary	0.5656	0.5387	0.5354
Blank-as-normal	0.5360	0.5548	0.5148

S4. Conditional component-resampling contrasts

Values are pooled OOF macro-F1 differences. The 2,000 shared bootstrap draws resample the 96 content components within five normal-only and 91 mixed-component strata. Intervals use the 2.5th and 97.5th percentiles; all draws are valid for every listed contrast. Fitted predictors and historical partitions remain fixed, so the intervals describe cohort-composition sensitivity. Fold-mean scores are reported separately in the article.

Table S3: Twenty-four specified main-model contrasts.

Policy	Model	Contrast	Estimate	Conditional95% interval
Primary	RF100	Full random − recording	+0.5827	[+0.4363,+0.6960]
Primary	RF100	Full random − temporal	+0.1798	[+0.1284,+0.2398]
Primary	RF100	Recording context − full	+0.1853	[+0.0812,+0.3154]
Primary	Extra Trees	Full random − recording	+0.5480	[+0.3917,+0.6679]
Primary	Extra Trees	Full random − temporal	+0.1973	[+0.1411,+0.2593]
Primary	Extra Trees	Recording context − full	+0.0047	[−0.1596,+0.1781]
Primary	HistGB	Full random − recording	+0.5249	[+0.3928,+0.6350]
Primary	HistGB	Full random − temporal	+0.1490	[+0.1099,+0.1991]
Primary	HistGB	Recording context − full	+0.1015	[−0.0648,+0.3390]
Primary	Logistic regression	Full random − recording	+0.3411	[+0.2129,+0.4169]
Primary	Logistic regression	Full random − temporal	+0.0424	[+0.0186,+0.0690]
Primary	Logistic regression	Recording context − full	−0.1297	[−0.2353,−0.0596]
Blank-as-normal	RF100	Full random − recording	+0.5455	[+0.4881,+0.6042]
Blank-as-normal	RF100	Full random − temporal	+0.3496	[+0.2820,+0.4257]
Blank-as-normal	RF100	Recording context − full	−0.0364	[−0.0893,+0.0164]
Blank-as-normal	Extra Trees	Full random − recording	+0.4760	[+0.4232,+0.5338]
Blank-as-normal	Extra Trees	Full random − temporal	+0.3346	[+0.2712,+0.4002]
Blank-as-normal	Extra Trees	Recording context − full	−0.0360	[−0.0809,+0.0072]
Blank-as-normal	HistGB	Full random − recording	+0.5303	[+0.4666,+0.5905]
Blank-as-normal	HistGB	Full random − temporal	+0.3107	[+0.2575,+0.3718]
Blank-as-normal	HistGB	Recording context − full	+0.0247	[−0.0119,+0.0666]

Policy	Model	Contrast	Estimate	Conditional95% interval
Blank-as-normal	Logistic regression	Full random − recording	+0.1197	[+0.0578,+0.1822]
Blank-as-normal	Logistic regression	Full random − temporal	+0.0444	[+0.0241,+0.0644]
Blank-as-normal	Logistic regression	Recording context − full	−0.0382	[−0.0852,+0.0071]

Table S5: Additional paired RF ablations using the same draws.

Policy	Contrast	Estimate	Conditional95% interval
Primary	Recording numeric-minus-context − full	−0.1179	[−0.2486,−0.0208]
Blank-as-normal	Recording numeric-minus-context − full	−0.0610	[−0.1064,−0.0185]

S5. Content-component and byte-copy-removal designs

H1 uses all 101 sources. Seed 42 evaluates RF100 and HGB with full, context and IMU features under both label policies (60 fit-folds). Seeds 43/44 add RF100 full/context checks (40 fit-folds). H2 removes byte-identical source 47 and evaluates RF100 across three representations, three splits and both policies (90 fit-folds). The H2 cohort retains the remaining partial inertial matches.

Across the twelve same-cohort H1 comparisons, six pooled changes are positive and six negative, approximately -0.0462 to $+0.0340$. H2 scores below concern its 100-file cohort. Ten policy/fold membership checks retain the original inclusive one-second temporal embargo. Independent verification covers 190 model hashes and 1,795,048 prediction records across configurations.

Table S6: H2 RF full-feature pooled OOF macro-F1, rounded to three decimals.

Policy	Cohort supervised rows	Random	Temporal	Recording
Primary	32,838	0.995	0.818	0.445
Blank-as-normal	61,484	0.931	0.583	0.391

S6. STRIDE recording-type analysis

The STRIDE v4 archive supplies nine bump and eight pothole recording folders from one POCO X2 device identifier (Khandakar et al. 2024, 2025a). A whole recording is one example. Acceleration and gyroscope vector norms are summarised separately by mean, standard deviation, RMS, range and maximum absolute value, giving ten features. Duration is evaluated as a separate input. Six driving-behaviour recordings are excluded from this label task.

Five-fold stratified evaluation uses seeds 42–44, an RF with 100 trees, depth 5 and minimum leaf size 2, and scaled logistic regression. The table reports pooled OOF scores across the same 17 recording types. The archive SHA-256 is `475853a7aaf9a15068dc9c17014f3e46197acc4b705cd3851b84692cdd82f851`; the source record states CC BY 4.0.

Table S7: Pooled OOF macro-F1 across the same 17 STRIDE observations.

Features	Model	Seed 42	Seed 43	Seed 44
Ten sensor norm summaries	RF100	0.7639	0.8235	0.8235
Ten sensor norm summaries	Scaled logistic regression	0.7639	0.8786	0.8211
Duration only	RF100	0.5882	0.5278	0.4706
Duration only	Scaled logistic regression	0.3561	0.4396	0.3529

S7. Exploratory temporal representations and annotation runs

Historical RF recording-held-out primary/sensitivity macro-F1 is 0.3953/0.3972 for the 60-dimensional raw-axis one-second control, 0.3751/0.3759 for a 20-dimensional physical representation, and 0.4015/0.3706 for its prefix-calibrated variant. The raw-axis control was

serialised under scikit-learn 1.5.2. These exploratory comparisons have both runtime and representation differences; the controlled same-runtime temporal results are given in S10.

The raw annotation audit identifies 149 pothole and 199 bump continuous labelled runs, all with retained rows. Secondary equal-run-weight row agreement under random/recording/temporal evaluation is approximately 0.952/0.247/0.597 for potholes and 0.957/0.839/0.728 for bumps. These summaries concern agreement within annotation runs; physical event boundaries and false-alarm rates were not measured.

S8. Implementation revision record

Main_A_v1 stopped at an RF reproduction mismatch under scikit-learn 1.5.2. A correct-runtime check reproduced 6,577/6,577 labels in the checked first fold, and all 120 A RF reproduction folds subsequently passed. STRIDE_v1 stopped before fitting with a manifest NameError; v2 supplies the completed analysis. The first unexecuted H implementation was amended to include the temporal embargo before H_v2 fitting. C_v1 was interrupted during scheduler handoff; complete C_v2 supplies the reported block. These partial attempts remain in the version history and are excluded from canonical completion counts.

The review records contain the compact inventories and source hashes. Public split/result views omit signal matrices, direct label/prediction arrays and coordinates; complete historical replay requires the separately obtained source inputs.

S9. Anchor-stride sensitivity

Strides 5 and 20 add 40 RF100 recording-held-out fits; stride 10 is the existing reference. Both policies and full/IMU representations retain the original file-fold mapping. Full-cadence reconstruction at stride 10 matches cached features, ordered labels, source identities and exact fit/test sequences. All 101 files match their frozen hashes. Reloaded predictions and independently recomputed metrics agree for all 40 new fits.

All strides retain the same source population and three-class definition. Density changes alter observed rows, capped training composition and cap binding. The table therefore reports recording-held-out sampling sensitivity at each density.

Table S8: All eight new sampling configurations; recording-heldout seed 42, RF100.

Stride	Policy	Features	Rows	Fold mean macro-F1	Pooled OOF macro-F1
5	Primary	Full numeric	65,774	0.4386	0.4323
5	Primary	IMU12	65,774	0.3379	0.3408
5	Blank-as-normal	Full numeric	123,411	0.3866	0.3726
5	Blank-as-normal	IMU12	123,411	0.3432	0.3402
20	Primary	Full numeric	16,440	0.4086	0.4049
20	Primary	IMU12	16,440	0.3386	0.3400
20	Blank-as-normal	Full numeric	30,889	0.3765	0.3648
20	Blank-as-normal	IMU12	30,889	0.3563	0.3492

S10. Temporal controls and runtime verification

One-second windows use the closed past-inclusive interval, finite timestamps and stable source-row ordering for tied timestamps. Membership matches a naive implementation in 13 synthetic cases and 303 real anchors across 101 sources. All 20 policy/split/fold membership arrays and fit caps pass checks. The original time-block anchor embargo gives zero within-source raw-window intersections; no additional anchors are removed.

The sixteen configurations contain eight matched feature comparisons. All 80 models reproduce their stored predictions after reload. The 40 instantaneous controls reproduce the original A results after row-identity alignment and class decoding. For the ten historical axis-window models, source memberships, retained columns and medians match the inspected definitions, but serialisation uses scikit-learn 1.5.2. Rerunning the unchanged implementation under 1.9.0 creates ten separate reference models; all ten RF/window recording cells reproduce their exact memberships and labels before the remaining cells are interpreted.

T_v1 stopped after two fits because of an object-array read error, and T_v2 stopped after two fits when the historical runtime discrepancy was detected. T_v3 uses Unicode prediction arrays and the same-runtime reference with the original scientific design. The completion inventory distinguishes these attempts from the canonical temporal block.

Table S9: All 16 temporal configurations. Fold means and pooled OOF are distinct summaries of the same predictions.

Policy	Split	Model	Features	Fold mean macro-F1	Pooled OOF macro-F1
Primary	Recording	RF100	IMU12	0.3410	0.3423
Primary	Recording	RF100	Window60	0.4072	0.4042
Primary	Recording	HistGB	IMU12	0.3379	0.3356
Primary	Recording	HistGB	Window60	0.4173	0.4161
Primary	Time-block	RF100	IMU12	0.4454	0.4498
Primary	Time-block	RF100	Window60	0.5929	0.5970
Primary	Time-block	HistGB	IMU12	0.4419	0.4458
Primary	Time-block	HistGB	Window60	0.6015	0.6054
Blank-as-normal	Recording	RF100	IMU12	0.3448	0.3422
Blank-as-normal	Recording	RF100	Window60	0.4100	0.4039
Blank-as-normal	Recording	HistGB	IMU12	0.3392	0.3373
Blank-as-normal	Recording	HistGB	Window60	0.4023	0.3996
Blank-as-normal	Time-block	RF100	IMU12	0.3591	0.3614
Blank-as-normal	Time-block	RF100	Window60	0.4256	0.4280
Blank-as-normal	Time-block	HistGB	IMU12	0.3581	0.3603
Blank-as-normal	Time-block	HistGB	Window60	0.4294	0.4317

Table S11: Temporal-extension completion and evidence inventory.

Targeted block	Canonical fit-folds	Independent checks
Anchor stride 5/20 sampling	40	8 configurations, all 40 reloaded predictions and model hashes verified
Same-runtime original-window reference	10	All 10 serialized model versions, parameters and hashes checked; predictions reproduced by T
T temporal diagnostic	80	16 configurations; all 80 source/label/model/prediction checks pass;40 controls match A
Targeted total	130	Distinct from the original 1,580-fit suite
Combined canonical total	1,710	Computational fits, not independent acquisitions

S11. PVS acquisition, candidate and selector results

PVS uses the released left-dashboard accelerometer/gyroscope streams and sample-index labels after exact length and column checks (Menegazzo and von Wangenheim n.d., 2023). The primary population contains 43,204 windows from 1,080,905 raw rows; the sensitivity contains 10,804 disjoint windows. For each grid, 240 inner fits and 24 outer refits yield 264 models. The three selectors share candidate-specific outer predictions.

Tables give acquisition counts, individual-domain candidate scores, selector outcomes, class recalls and confusion counts. Class order is 0=no speed bump and 1=speed bump, with TN/FP/FN/TP in that order. Metrics use four decimals and confusion counts are exact integers. Axis summaries average three outer domains from the nine acquisitions. Regret compares the selected score with the best outer score among the same four candidates as a descriptive reference.

Independent checks reload all 528 initial PVS models, reconstruct split/cap memberships and medians, and recompute predictions, confusion matrices and selector summaries. Feature checks recompute 315 deterministic windows and verify retained anchor labels. The primary two-axis MAE criterion is 0.02; its outcome and the disjoint-window sensitivity are reported in the article.

Table S12: PVS source population by trip

Trip	Vehicle/driver	Scenario	Raw rows	Windows	Speed-bump windows
PVS 1	Volkswagen Saveiro / Driver 1	1	144036	5758	138
PVS 2	Volkswagen Saveiro / Driver 1	2	124684	4983	79
PVS 3	Volkswagen Saveiro / Driver 1	3	105816	4229	24
PVS 4	Fiat Bravo / Driver 2	1	132492	5296	124
PVS 5	Fiat Bravo / Driver 2	2	133877	5352	77
PVS 6	Fiat Bravo / Driver 2	3	96279	3848	25
PVS 7	Fiat Palio / Driver 3	1	128548	5138	116
PVS 8	Fiat Palio / Driver 3	2	123618	4941	74
PVS 9	Fiat Palio / Driver 3	3	91555	3659	25

Table S13: Primary PVS candidate metrics by vehicle/driver domain

Outer domain	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
vehicle_1	RF100 IMU	0.8639	0.8137	0.7974	0.8599	0.7983
vehicle_1	RF100 IMU+context	0.8972	0.8332	0.8124	0.8543	0.8001
vehicle_1	HGB IMU	0.8970	0.8391	0.8187	0.8641	0.8025
vehicle_1	HGB IMU+context	0.9217	0.8664	0.8618	0.8492	0.7858
vehicle_2	RF100 IMU	0.8670	0.8131	0.7356	0.8155	0.8499
vehicle_2	RF100 IMU+context	0.8763	0.8059	0.7726	0.8625	0.8823
vehicle_2	HGB IMU	0.8968	0.8397	0.7652	0.8494	0.8859
vehicle_2	HGB IMU+context	0.9144	0.8470	0.7985	0.9049	0.9164
vehicle_3	RF100 IMU	0.8731	0.8168	0.7884	0.7532	0.9253
vehicle_3	RF100 IMU+context	0.8964	0.8419	0.8349	0.7887	0.9262
vehicle_3	HGB IMU	0.9086	0.8500	0.8319	0.7917	0.9421
vehicle_3	HGB IMU+context	0.9268	0.8659	0.8352	0.8497	0.9459

Table S14: Primary PVS candidate metrics by scenario domain

Outer domain	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
scenario_1	RF100 IMU	0.9147	0.8757	0.8574	0.7186	0.6623
scenario_1	RF100 IMU+context	0.9248	0.8931	0.6568	0.6824	0.6246
scenario_1	HGB IMU	0.9268	0.9026	0.8474	0.7152	0.6536
scenario_1	HGB IMU+context	0.9461	0.9095	0.6701	0.6702	0.6129
scenario_2	RF100 IMU	0.8401	0.7758	0.7368	0.8781	0.9210
scenario_2	RF100 IMU+context	0.8744	0.8057	0.6806	0.6318	0.5920
scenario_2	HGB IMU	0.8891	0.8043	0.7513	0.9026	0.9241
scenario_2	HGB IMU+context	0.9280	0.8499	0.6851	0.7074	0.6470
scenario_3	RF100 IMU	0.8612	0.8209	0.7697	0.8866	0.8711
scenario_3	RF100 IMU+context	0.8823	0.8339	0.7466	0.8588	0.8973
scenario_3	HGB IMU	0.9018	0.8498	0.7464	0.8845	0.8510
scenario_3	HGB IMU+context	0.9265	0.8792	0.7201	0.9143	0.8782

Table S15: Primary PVS selector metrics by vehicle/driver domain

Outer domain	Selector	Selected candidate	Inner estimate	Outer F1	Absolute error	Oracle regret
vehicle_1	Random	HGB IMU+context	0.9217	0.8492	0.0725	0.0149
vehicle_1	Trip	HGB IMU+context	0.8664	0.8492	0.0172	0.0149
vehicle_1	Matched	HGB IMU+context	0.8618	0.8492	0.0126	0.0149
vehicle_2	Random	HGB IMU+context	0.9144	0.9049	0.0095	0.0000
vehicle_2	Trip	HGB IMU+context	0.8470	0.9049	0.0579	0.0000
vehicle_2	Matched	HGB IMU+context	0.7985	0.9049	0.1064	0.0000
vehicle_3	Random	HGB IMU+context	0.9268	0.8497	0.0771	0.0000
vehicle_3	Trip	HGB IMU+context	0.8659	0.8497	0.0162	0.0000
vehicle_3	Matched	HGB IMU+context	0.8352	0.8497	0.0145	0.0000

Table S16: Primary PVS selector metrics by scenario domain

Outer domain	Selector	Selected candidate	Inner estimate	Outer F1	Absolute error	Oracle regret
scenario_1	Random	HGB IMU+context	0.9461	0.6702	0.2760	0.0484
scenario_1	Trip	HGB IMU+context	0.9095	0.6702	0.2393	0.0484
scenario_1	Matched	RF100 IMU	0.8574	0.7186	0.1388	0.0000
scenario_2	Random	HGB IMU+context	0.9280	0.7074	0.2206	0.1951
scenario_2	Trip	HGB IMU+context	0.8499	0.7074	0.1424	0.1951
scenario_2	Matched	HGB IMU	0.7513	0.9026	0.1513	0.0000
scenario_3	Random	HGB IMU+context	0.9265	0.9143	0.0122	0.0000
scenario_3	Trip	HGB IMU+context	0.8792	0.9143	0.0351	0.0000
scenario_3	Matched	RF100 IMU	0.7697	0.8866	0.1169	0.0277

Table S17: Non-overlap sensitivity candidate performance by held-out axis

Axis	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
Scenario	HGB IMU	0.8791	0.8512	0.7926	0.8168	0.7914
Scenario	HGB IMU+context	0.8851	0.8561	0.6804	0.7986	0.7522
Scenario	RF100 IMU	0.8217	0.8006	0.7606	0.7872	0.8134
Scenario	RF100 IMU+context	0.8497	0.8293	0.7229	0.7862	0.7924
Vehicle	HGB IMU	0.8578	0.8301	0.7995	0.8282	0.8539
Vehicle	HGB IMU+context	0.8669	0.8446	0.8217	0.8485	0.8587
Vehicle	RF100 IMU	0.8040	0.7825	0.7538	0.7871	0.8611
Vehicle	RF100 IMU+context	0.8336	0.8073	0.7818	0.8151	0.8774

Table S18: Non-overlap sensitivity selector estimation summary

Axis	Selector	Mean estimate	Mean outer F1	MAE	Outer-minus-inner bias
Scenario	Trip	0.8608	0.7954	0.1061	-0.0655
Scenario	Matched	0.7926	0.8168	0.1290	0.0242
Scenario	Random	0.8869	0.7904	0.1069	-0.0965
Vehicle	Trip	0.8446	0.8485	0.0497	0.0038
Vehicle	Matched	0.8240	0.8522	0.0535	0.0282
Vehicle	Random	0.8672	0.8380	0.0489	-0.0291

Table S19: Primary PVS class recalls and confusion counts by vehicle/driver domain

Outer domain	Candidate	No-bump recall	Bump recall	TN	FP	FN	TP
vehicle_1	RF100 IMU	0.9991	0.5975	14716	13	97	144
vehicle_1	RF100 IMU+context	0.9986	0.6017	14708	21	96	145
vehicle_1	HGB IMU	0.9992	0.6058	14717	12	95	146
vehicle_1	HGB IMU+context	0.9990	0.5726	14715	14	103	138
vehicle_2	RF100 IMU	0.9919	0.7080	14154	116	66	160
vehicle_2	RF100 IMU+context	0.9946	0.7699	14193	77	52	174
vehicle_2	HGB IMU	0.9931	0.7788	14172	98	50	176
vehicle_2	HGB IMU+context	0.9965	0.8363	14220	50	37	189
vehicle_3	RF100 IMU	0.9763	0.8744	13202	321	27	188
vehicle_3	RF100 IMU+context	0.9826	0.8698	13288	235	28	187
vehicle_3	HGB IMU	0.9819	0.9023	13278	245	21	194
vehicle_3	HGB IMU+context	0.9896	0.9023	13382	141	21	194

Table S20: Primary PVS class recalls and confusion counts by scenario domain

Outer domain	Candidate	No-bump recall	Bump recall	TN	FP	FN	TP
scenario_1	RF100 IMU	0.9966	0.3280	15761	53	254	124
scenario_1	RF100 IMU+context	0.9979	0.2513	15780	34	283	95
scenario_1	HGB IMU	0.9977	0.3095	15777	37	261	117
scenario_1	HGB IMU+context	0.9983	0.2275	15787	27	292	86
scenario_2	RF100 IMU	0.9942	0.8478	14958	88	35	195
scenario_2	RF100 IMU+context	0.9971	0.1870	15002	44	187	43
scenario_2	HGB IMU	0.9961	0.8522	14987	59	34	196
scenario_2	HGB IMU+context	0.9983	0.2957	15021	25	162	68
scenario_3	RF100 IMU	0.9989	0.7432	11649	13	19	55
scenario_3	RF100 IMU+context	0.9973	0.7973	11631	31	15	59
scenario_3	HGB IMU	0.9992	0.7027	11653	9	22	52
scenario_3	HGB IMU+context	0.9996	0.7568	11657	5	18	56

S12. Post hoc PVS coordinate and speed blocks

The follow-up retains parent populations, window grids, outer splits, capped fit rows, settings and training-only median imputation. It adds 48 fits and reuses 48 parent predictions, giving 96 domain-level model outcomes. Inputs are IMU (30), IMU+speed (31), IMU+latitude/longitude (32), and their combination (33 fields). Model inputs are float32. Latitude/longitude form the coordinate block; the separate speed field is GPS-derived in the source.

The five contrasts are speed minus IMU, coordinates minus IMU, combined context minus IMU, coordinates conditional on speed, and speed conditional on coordinates. Across eight grid/axis/classifier means, their ranges and positive/negative counts are respectively: -0.0157 to $+0.0212$ (4/4); -0.0239 to $+0.0275$ (3/5); -0.1034 to $+0.0329$ (4/4); -0.0898 to $+0.0317$ (5/3); and -0.0795 to $+0.0239$ (5/3). No contrast mean is exactly zero. The tables give every do-

main result and signed difference. These comparisons describe retrained feature representations within the inspected acquisitions.

Table S21: Domain-mean performance for the primary grid.

Axis	Model	Features	Macro-F1	Balanced accuracy
Vehicle	RF100	IMU	0.8095	0.8579
Vehicle	RF100	+speed	0.8305	0.8665
Vehicle	RF100	+GPS	0.8263	0.8642
Vehicle	RF100	+GPS+speed	0.8352	0.8695
Vehicle	HGB	IMU	0.8351	0.8768
Vehicle	HGB	+speed	0.8562	0.8839
Vehicle	HGB	+GPS	0.8625	0.8868
Vehicle	HGB	+GPS+speed	0.8679	0.8827
Scenario	RF100	IMU	0.8278	0.8181
Scenario	RF100	+speed	0.8141	0.7996
Scenario	RF100	+GPS	0.8038	0.7765
Scenario	RF100	+GPS+speed	0.7243	0.7046
Scenario	HGB	IMU	0.8341	0.8096
Scenario	HGB	+speed	0.8356	0.8027
Scenario	HGB	+GPS	0.8102	0.7607
Scenario	HGB	+GPS+speed	0.7640	0.7127

Table S23: Domain-mean performance for the nonoverlap grid.

Axis	Model	Features	Macro-F1	Balanced accuracy
Vehicle	RF100	IMU	0.7871	0.8611
Vehicle	RF100	+speed	0.7926	0.8560
Vehicle	RF100	+GPS	0.8022	0.8676
Vehicle	RF100	+GPS+speed	0.8151	0.8774
Vehicle	HGB	IMU	0.8282	0.8539
Vehicle	HGB	+speed	0.8168	0.8418
Vehicle	HGB	+GPS	0.8246	0.8498
Vehicle	HGB	+GPS+speed	0.8485	0.8587
Scenario	RF100	IMU	0.7872	0.8134
Scenario	RF100	+speed	0.7715	0.7950
Scenario	RF100	+GPS	0.7789	0.7973
Scenario	RF100	+GPS+speed	0.7862	0.7924
Scenario	HGB	IMU	0.8168	0.7914
Scenario	HGB	+speed	0.8146	0.7938

Axis	Model	Features	Macro-F1	Balanced accuracy
Scenario	HGB	+GPS	0.8074	0.7698
Scenario	HGB	+GPS+speed	0.7986	0.7522

Table S25: Primary vehicle-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	+0.0209	+0.0124	+0.0262	+0.0242
RF100	GPS − IMU	+0.0167	-0.0021	+0.0177	+0.0346
RF100	GPS+speed − IMU	+0.0256	-0.0057	+0.0470	+0.0355
RF100	GPS given speed	+0.0047	-0.0181	+0.0208	+0.0113
RF100	Speed given GPS	+0.0089	-0.0036	+0.0293	+0.0009
HGB	Speed − IMU	+0.0212	+0.0159	+0.0211	+0.0266
HGB	GPS − IMU	+0.0275	-0.0091	+0.0429	+0.0486
HGB	GPS+speed − IMU	+0.0329	-0.0149	+0.0555	+0.0580
HGB	GPS given speed	+0.0117	-0.0308	+0.0344	+0.0315
HGB	Speed given GPS	+0.0054	-0.0059	+0.0126	+0.0095

Table S26: Primary scenario-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	-0.0137	-0.0097	-0.0341	+0.0028
RF100	GPS − IMU	-0.0239	-0.0326	-0.0344	-0.0048
RF100	GPS+speed − IMU	-0.1034	-0.0362	-0.2462	-0.0279
RF100	GPS given speed	-0.0898	-0.0265	-0.2121	-0.0306
RF100	Speed given GPS	-0.0795	-0.0036	-0.2119	-0.0230
HGB	Speed − IMU	+0.0014	+0.0059	-0.0242	+0.0227
HGB	GPS − IMU	-0.0239	-0.0449	-0.0572	+0.0304
HGB	GPS+speed − IMU	-0.0701	-0.0451	-0.1951	+0.0298
HGB	GPS given speed	-0.0716	-0.0510	-0.1709	+0.0071
HGB	Speed given GPS	-0.0462	-0.0002	-0.1379	-0.0006

Table S27: Nonoverlap vehicle-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	+0.0055	-0.0135	+0.0261	+0.0040
RF100	GPS − IMU	+0.0151	+0.0090	+0.0231	+0.0133
RF100	GPS+speed − IMU	+0.0280	-0.0048	+0.0628	+0.0259
RF100	GPS given speed	+0.0225	+0.0087	+0.0367	+0.0219
RF100	Speed given GPS	+0.0129	-0.0138	+0.0398	+0.0126
HGB	Speed − IMU	-0.0114	-0.0135	-0.0168	-0.0041
HGB	GPS − IMU	-0.0036	-0.0410	+0.0167	+0.0135
HGB	GPS+speed − IMU	+0.0202	-0.0112	+0.0406	+0.0313
HGB	GPS given speed	+0.0317	+0.0023	+0.0573	+0.0354
HGB	Speed given GPS	+0.0239	+0.0298	+0.0239	+0.0178

Table S28: Nonoverlap scenario-domain macro-F1 contrasts.

Model	Contrast	Mean gain	Domain 1	Domain 2	Domain 3
RF100	Speed − IMU	-0.0157	-0.0123	-0.0349	+0.0000
RF100	GPS − IMU	-0.0083	-0.0161	+0.0069	-0.0158
RF100	GPS+speed − IMU	-0.0010	-0.0070	+0.0079	-0.0040
RF100	GPS given speed	+0.0147	+0.0054	+0.0428	-0.0040
RF100	Speed given GPS	+0.0073	+0.0092	+0.0010	+0.0118
HGB	Speed − IMU	-0.0021	+0.0470	-0.0089	-0.0446
HGB	GPS − IMU	-0.0094	+0.0018	-0.0063	-0.0236
HGB	GPS+speed − IMU	-0.0181	+0.0097	-0.0889	+0.0248
HGB	GPS given speed	-0.0160	-0.0374	-0.0800	+0.0693
HGB	Speed given GPS	-0.0088	+0.0079	-0.0826	+0.0484

Table S29: Vehicle-minus-scenario differences in domain-mean gains.

Grid	Model	Speed − IMU	GPS − IMU	All − IMU	GPS given speed	Speed given GPS
Primary	RF100	+0.0346	+0.0407	+0.1290	+0.0944	+0.0884
Primary	HGB	+0.0197	+0.0514	+0.1030	+0.0833	+0.0516
Nonoverlap	RF100	+0.0213	+0.0235	+0.0290	+0.0077	+0.0055
Nonoverlap	HGB	-0.0093	+0.0058	+0.0384	+0.0477	+0.0326

S13. Common-budget PVS diagnostic

The post hoc diagnostic adds 264 fits over the existing disjoint-window population. Common per-class budgets are set by the minimum support across inner and outer training pools in each outer domain. Validation/test observations, candidate settings and saved acquisition groups are retained. V denotes vehicle/driver and S scenario; candidate suffix S denotes sensor features and C adds coordinates and speed. Signed bias is outer minus inner F1.

The tables separately report equally budgeted outer refits and historical-full-development outer predictions for the newly selected candidates. With context and budget diagnostics, the computational inventory contains 840 PVS fits and 2,550 fits overall before Mexico. The scientific grouping remains nine PVS acquisitions. Future confirmation should use new whole acquisitions after development choices are frozen, retaining dates, crossed vehicle/driver identities and shared route/event identifiers.

Table S30: Common per-class training budgets by outer domain.

Axis	Held-out domain	Class 0	Class 1
Vehicle	1	3383	53
Vehicle	2	3383	53
Vehicle	3	3568	56
Scenario	1	2917	18
Scenario	2	2917	18
Scenario	3	3763	57

Table S31: Common-budget selected-system domain outcomes.

Axis	ID	Selector	Model	Inner F1	Outer F1	Bias	Abs. error
V	1	Random	HGB/S+C	0.8684	0.8444	-0.0239	0.0239
V	1	Trip	HGB/S+C	0.8644	0.8444	-0.0200	0.0200
V	1	Matched	HGB/S	0.8383	0.8618	+0.0236	0.0236
V	2	Random	HGB/S+C	0.8549	0.8590	+0.0041	0.0041
V	2	Trip	HGB/S+C	0.8166	0.8590	+0.0425	0.0425
V	2	Matched	HGB/S+C	0.8069	0.8590	+0.0521	0.0521
V	3	Random	HGB/S+C	0.8510	0.8371	-0.0139	0.0139
V	3	Trip	HGB/S+C	0.8398	0.8371	-0.0027	0.0027
V	3	Matched	HGB/S+C	0.8415	0.8371	-0.0044	0.0044
S	1	Random	HGB/S+C	0.8960	0.6873	-0.2087	0.2087
S	1	Trip	RF/S+C	0.8919	0.7048	-0.1872	0.1872
S	1	Matched	HGB/S	0.8501	0.6842	-0.1659	0.1659
S	2	Random	HGB/S+C	0.8365	0.5588	-0.2777	0.2777
S	2	Trip	HGB/S+C	0.7730	0.5588	-0.2142	0.2142
S	2	Matched	HGB/S	0.7557	0.5447	-0.2110	0.2110
S	3	Random	HGB/S+C	0.8770	0.8787	+0.0017	0.0017
S	3	Trip	HGB/S+C	0.8517	0.8787	+0.0270	0.0270
S	3	Matched	HGB/S	0.7163	0.8565	+0.1402	0.1402

Table S32: Historical-full-budget selected-system domain outcomes.

Axis	ID	Selector	Model	Inner F1	Outer F1	Bias	Abs. error
V	1	Random	HGB/S+C	0.8684	0.8129	-0.0554	0.0554
V	1	Trip	HGB/S+C	0.8644	0.8129	-0.0515	0.0515
V	1	Matched	HGB/S	0.8383	0.8241	-0.0142	0.0142
V	2	Random	HGB/S+C	0.8549	0.9004	+0.0455	0.0455
V	2	Trip	HGB/S+C	0.8166	0.9004	+0.0838	0.0838
V	2	Matched	HGB/S+C	0.8069	0.9004	+0.0935	0.0935
V	3	Random	HGB/S+C	0.8510	0.8320	-0.0190	0.0190
V	3	Trip	HGB/S+C	0.8398	0.8320	-0.0077	0.0077
V	3	Matched	HGB/S+C	0.8415	0.8320	-0.0094	0.0094
S	1	Random	HGB/S+C	0.8960	0.7046	-0.1914	0.1914
S	1	Trip	RF/S+C	0.8919	0.7280	-0.1640	0.1640
S	1	Matched	HGB/S	0.8501	0.6949	-0.1552	0.1552
S	2	Random	HGB/S+C	0.8365	0.7800	-0.0566	0.0566
S	2	Trip	HGB/S+C	0.7730	0.7800	+0.0069	0.0069
S	2	Matched	HGB/S	0.7557	0.8688	+0.1131	0.1131
S	3	Random	HGB/S+C	0.8770	0.9113	+0.0342	0.0342
S	3	Trip	HGB/S+C	0.8517	0.9113	+0.0596	0.0596
S	3	Matched	HGB/S	0.7163	0.8865	+0.1702	0.1702

S14. Mexico/Puebla released-subset analysis and source screening

The Applied Sciences supplement and related Vehicles description refer to one collection (Aguilar-González and Medina Santiago 2025, 2026). The four released arrays contain 36,000 positions, while the reported complete population exceeds 360,000; their subset/version mapping is unresolved. Every event-label run has six positions for potholes or eleven for bumps/braking. The original 50-position majority proposal retained no pothole/bump windows and was replaced before fitting. The retained protocol fixes ten-position disjoint windows, majority support of at least six, and exclusion of braking-containing windows. Blank denotes the release's non-event class.

Only acceleration x/y/z is used because the additional vibration channels are described as derived from acceleration. Each of x/y/z and vector magnitude supplies mean, population SD, minimum, maximum, RMS, 25th/50th/75th percentiles and first-difference population SD, giving 36 features. No exact six-channel row or consecutive 2/5/10/25-position fragment is shared between files, and all values are finite. The source files provide no timestamps, GPS or vehicle/driver identifiers.

Outer folds hold out files 1+2 (urban) or 3+4 (suburban), using documentation-derived route grouping. Random-window and whole-trip two-fold selectors operate within the single development family. Per-class budgets are (484,67,80) with urban held out and (444,48,93) with suburban held out. Full refits use 1,305 or 1,199 windows; common-budget refits use 631 or 585. RF100/HGB parameters match the fixed settings in the article. Selection uses pooled three-class OOF macro-F1, with RF first on exact ties. The 24 fits comprise 16 inner and eight outer fits.

The whole-trip-minus-random outer-F1 difference is zero under both targets. Mean absolute-error differences are -0.0008369747 for full-development and -0.0007920153 for common-budget refits, across two route-specific pairs. Independent reconstruction checks all 3,600 inclusion decisions, 2,504 retained feature/label records, split and class-budget memberships, 24 model hashes, reloaded predictions, imputation medians and metrics. Feature reconstruction agrees within 10^{-10}.

S14.1. Indonesia label-version screening

The Indonesia author archive (Setiawan et al. 2021), obtained on 26 October 2026, contains 60 CSVs, including concurrent phone streams, chunks and alternate exports. Source screening identifies six motorcycle session families and ten timestamped car-session candidates. Five base motorcycle files overlap wider six-class exports. For matched Samsung records, original labels agree with wider column 21 on 20,782/25,833 rows in M5-1 and 4,456/5,206 in M5-4; agreement with column 22 is 22,558/25,833 and 4,693/5,206. Wider class codes and processed/LSTM-export fields lack a documented mapping in the base README. This source remains at the label-clarification stage, with no predictive fits included.

Table S33: Retained nonoverlap window counts; source files have 9,000 positions each.

Source	Non-event	Pothole	Bump	Total
Trip 1	454	48	96	598
Trip 2	444	64	93	601
Trip 3	484	67	80	631
Trip 4	516	78	80	674
Total	1,898	257	349	2,504

Table S34: Mexico task: primary full-development refit.

Held-out route	Inner selector	Chosen	Inner F1	Outer F1	Abs. error
Urban (1+2)	Random	RF	0.9922	0.9934	0.0012
Urban (1+2)	Whole-trip	RF	0.9922	0.9934	0.0012
Suburban (3+4)	Random	RF	0.9934	0.9960	0.0026
Suburban (3+4)	Whole-trip	RF	0.9950	0.9960	0.0010

Table S35: Mexico task: separately specified common-budget outer sensitivity.

Held-out route	Inner selector	Chosen	Inner F1	Outer F1	Abs. error
Urban (1+2)	Random	RF	0.9922	0.9918	0.0004
Urban (1+2)	Whole-trip	RF	0.9922	0.9918	0.0004
Suburban (3+4)	Random	RF	0.9934	0.9960	0.0026
Suburban (3+4)	Whole-trip	RF	0.9950	0.9960	0.0010

S15. Additional RoadSens and PVS controls

Table S36: Fixed classifier parameters.

Model	Frozen main settings
Random forest (RF100)	100 trees; maximum depth 20; minimum leaf size 5; square-root feature sampling; balanced class weights
Extra Trees	100 trees; maximum depth 20; minimum leaf size 5; square-root feature sampling; balanced class weights; no bootstrap
Histogram gradient boosting (HistGB)	learning rate 0.1; 100 iterations; at most 31 leaves; minimum leaf size 20; L2 regularization 1; balanced class weights; early stopping disabled
Logistic regression	training scaling; C=1; L-BFGS; at most 2,000 iterations; tolerance 10^{-4}; balanced class weights

Table S37: RoadSens forest-capacity and fit-budget sensitivity; five-fold mean macro-F1.

Policy	Forest / cap	Random	Temporal	Recording
Primary	RF100 / 20,000	0.9945	0.8095	0.4289
Primary	RF100 / uncapped	0.9964	0.8168	0.4307
Primary	RF300 / 20,000	0.9969	0.8274	0.4774
Primary	RF300 / uncapped	0.9984	0.8300	0.4502
Blank-as-normal	RF100 / 20,000	0.9314	0.5775	0.3954
Blank-as-normal	RF100 / uncapped	0.9703	0.5931	0.3880
Blank-as-normal	RF300 / 20,000	0.9572	0.5908	0.3847
Blank-as-normal	RF300 / uncapped	0.9824	0.5978	0.3788

Table S38: Explicit-anomaly buffered spatial diagnostics; five-fold mean macro-F1.

Features	Model	100-m cells	500-m cells
Full numeric	RF100	0.6349	0.6215
Full numeric	HistGB	0.6551	0.6714
Context	RF100	0.6450	0.6702
Context	HistGB	0.6806	0.7277
Instantaneous IMU	RF100	0.5150	0.4989
Instantaneous IMU	HistGB	0.5125	0.4979

Table S39: RoadSens recording-held-out anchor-stride sensitivity; five-fold mean macro-F1.

Policy	Source-row stride	Supervised rows	Full numeric	Instantaneous IMU
Primary	5	65,774	0.4386	0.3379
Primary	10 (reference)	32,885	0.4289	0.3410
Primary	20	16,440	0.4086	0.3386
Blank-as-normal	5	123,411	0.3866	0.3432
Blank-as-normal	10 (reference)	61,730	0.3954	0.3448
Blank-as-normal	20	30,889	0.3765	0.3563

Table S40: Primary PVS candidate means by held-out axis.

Axis	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
Scenario	HGB IMU	0.9059	0.8522	0.7817	0.8341	0.8096
Scenario	HGB IMU+context	0.9336	0.8795	0.6918	0.7640	0.7127
Scenario	RF100 IMU	0.8720	0.8241	0.7880	0.8278	0.8181
Scenario	RF100 IMU+context	0.8939	0.8442	0.6947	0.7243	0.7046
Vehicle	HGB IMU	0.9008	0.8429	0.8053	0.8351	0.8768
Vehicle	HGB IMU+context	0.9210	0.8598	0.8318	0.8679	0.8827
Vehicle	RF100 IMU	0.8680	0.8146	0.7738	0.8095	0.8579
Vehicle	RF100 IMU+context	0.8900	0.8270	0.8066	0.8352	0.8695

Table S41: PVS context-block macro-F1 by window grid, held-out axis and classifier.

Grid	Axis	Model	IMU	+speed	+GPS	+GPS+speed
Primary	Vehicle	RF100	0.8095	0.8305	0.8263	0.8352
Primary	Vehicle	HGB	0.8351	0.8562	0.8625	0.8679
Primary	Scenario	RF100	0.8278	0.8141	0.8038	0.7243
Primary	Scenario	HGB	0.8341	0.8356	0.8102	0.7640
Nonoverlap	Vehicle	RF100	0.7871	0.7926	0.8022	0.8151
Nonoverlap	Vehicle	HGB	0.8282	0.8168	0.8246	0.8485
Nonoverlap	Scenario	RF100	0.7872	0.7715	0.7789	0.7862
Nonoverlap	Scenario	HGB	0.8168	0.8146	0.8074	0.7986

Supplementary figures

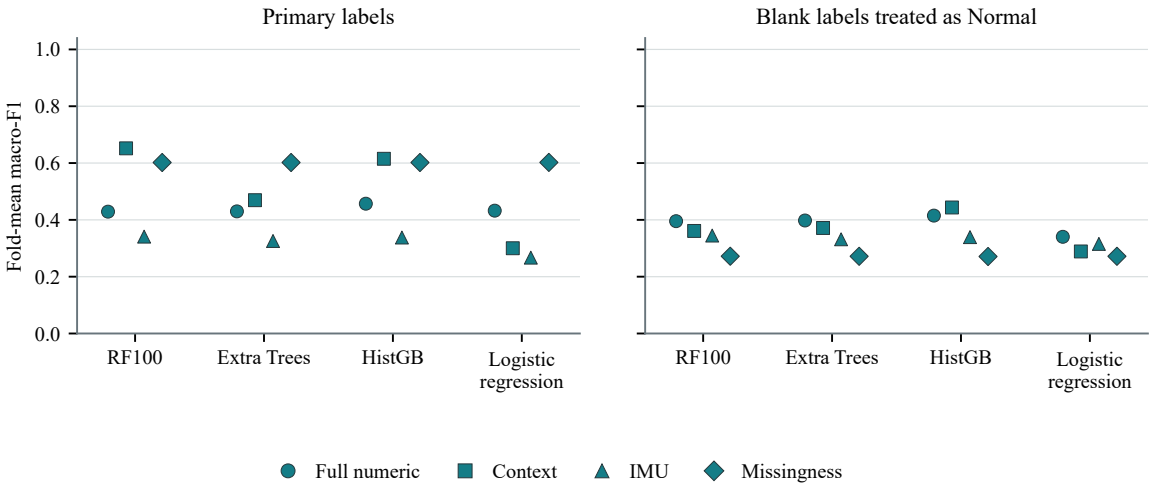


Figure S1: Recording-held-out feature comparisons. Points show five-fold mean macro-F1.

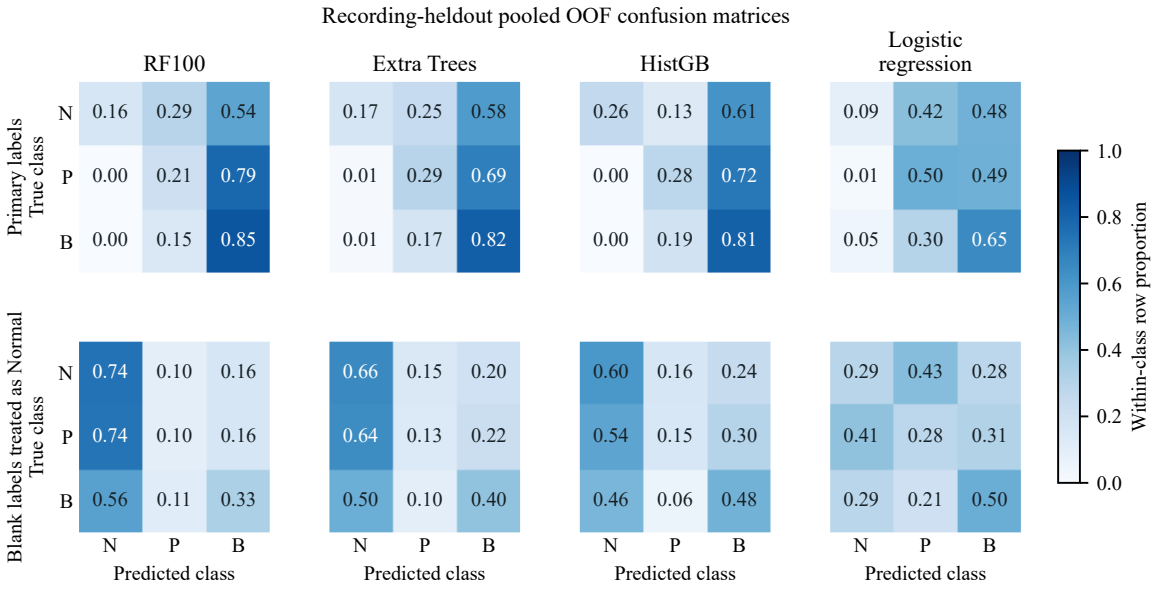


Figure S2: Full-feature recording-held-out pooled confusion matrices, normalised within true class. Diagonal values are recall; N=Normal, P=Pothole, B=Bump.

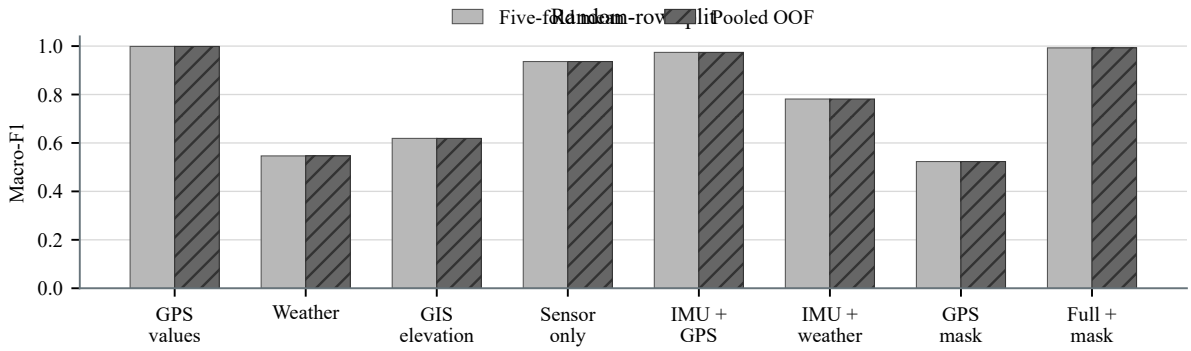


Figure S3: RF ablations under the primary policy and random-row evaluation. Bars compare fold-mean and pooled OOF macro-F1.

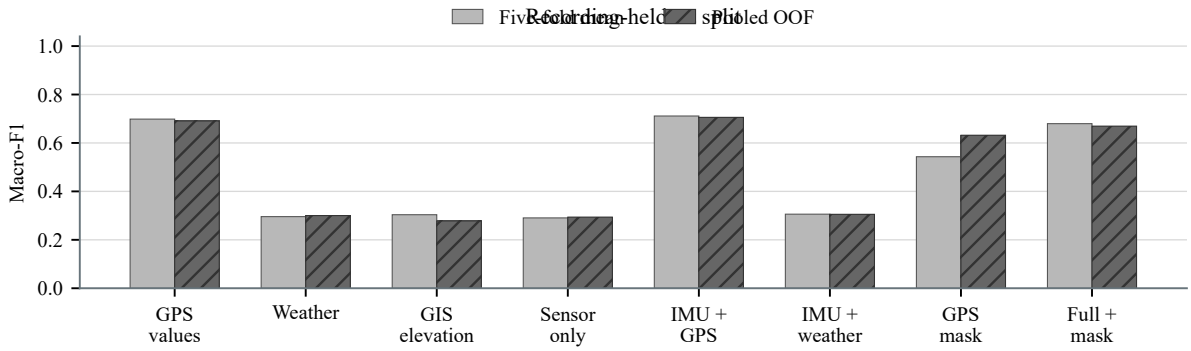


Figure S4: RF ablations under the primary policy and recording holdout. Bars compare fold-mean and pooled OOF macro-F1.

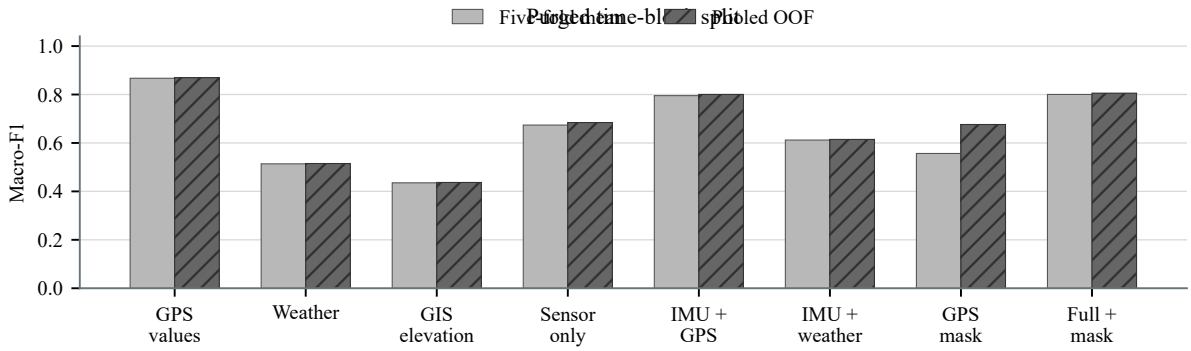


Figure S5: RF ablations under the primary policy and purged time blocks. Bars compare fold-mean and pooled OOF macro-F1.

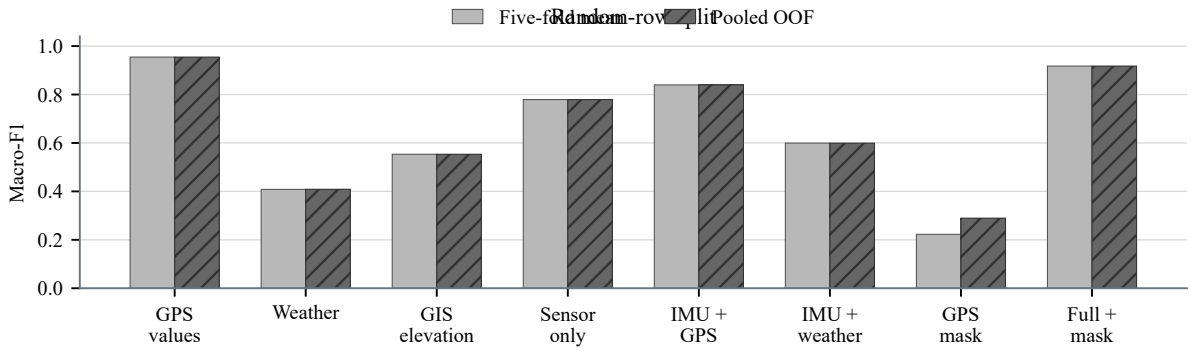


Figure S6: RF ablations under blank-as-normal and random-row evaluation. Bars compare fold-mean and pooled OOF macro-F1.

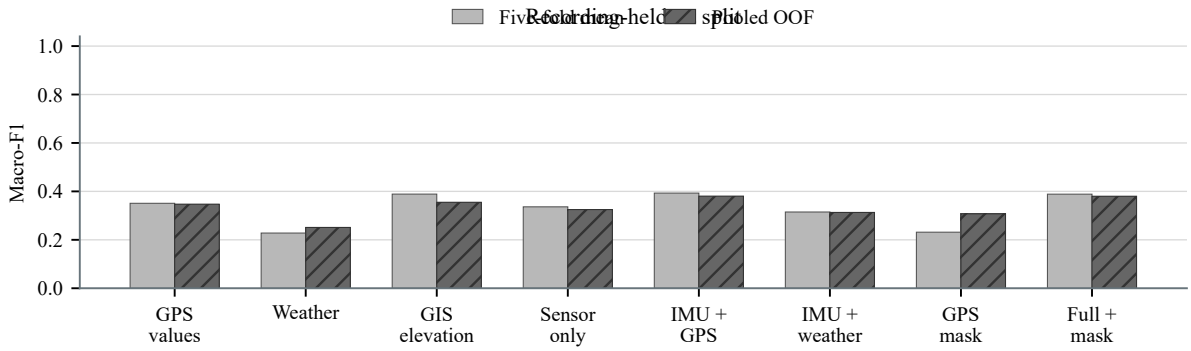


Figure S7: RF ablations under blank-as-normal and recording holdout. Bars compare fold-mean and pooled OOF macro-F1.

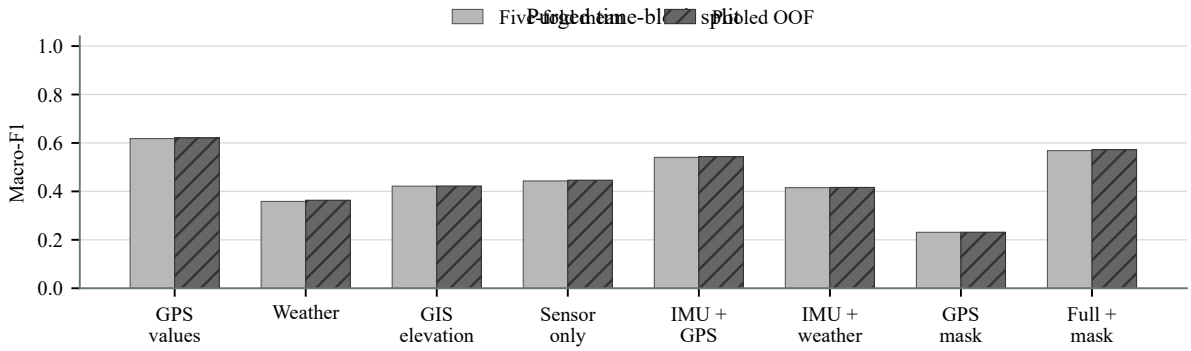


Figure S8: RF ablations under blank-as-normal and purged time blocks. Bars compare fold-mean and pooled OOF macro-F1.

S16. PVS annotation-interval engineering diagnostic

The protocol `PVS-ANNOTATION-EVENT-20261103-v1` was fixed after the original classification analysis and before event outcomes were computed. It reuses frozen outer predictions. The original protocol SHA-256 is `2e56a31b564da2d1cb2ba669c23b4c753daf86fb613d2813ab21c63796ad6d05`. The 60 contiguous positive source-label intervals are passages in the existing nine acquisitions. Every interval contains an assessed decision anchor. Negative exposure totals 2.9522222 h per axis and selector. A 100-row trailing window supplies a hard decision every 25 rows; a non-adjacent grid position or a negative decision separates alarm clusters.

An eligible alarm/event edge requires a positive decision timestamp in the annotated interval, expanded by the stated tolerance. Deterministic maximum-cardinality one-to-one matching counts hits. An alarm with no eligible edge is a non-event alarm; an unmatched alarm with an edge is a redundant event-associated notification. The initial calculation retained all unmatched alarms as a notification-burden endpoint. After inspecting that output, reporting amendment A1 separated its two components without changing predictions, thresholds, boundaries, matching, exposure or original counts. Both unmatched totals and the clarification are retained in the analysis records. Offsets are relative to annotation starts and describe annotation timing, with no physical-impact latency interpretation.

Table S42: PVS per-acquisition engineering counts at zero tolerance.

Axis	Trip	Selector	Hits/intervals	Non-event	Redundant	Negative h
Vehicle	1	Random	10/11	3	3	0.3902
Vehicle	2	Random	7/7	8	1	0.3405
Vehicle	3	Random	2/2	0	1	0.2919
Vehicle	1	Trip	10/11	3	3	0.3902
Vehicle	2	Trip	7/7	8	1	0.3405
Vehicle	3	Trip	2/2	0	1	0.2919
Vehicle	1	Matched	10/11	3	3	0.3902
Vehicle	2	Matched	7/7	8	1	0.3405
Vehicle	3	Matched	2/2	0	1	0.2919
Vehicle	4	Random	11/11	13	2	0.3590
Vehicle	5	Random	7/7	4	3	0.3662
Vehicle	6	Random	2/2	1	0	0.2655
Vehicle	4	Trip	11/11	13	2	0.3590

Axis	Trip	Selector	Hits/intervals	Non-event	Redundant	Negative h
Vehicle	5	Trip	7/7	4	3	0.3662
Vehicle	6	Trip	2/2	1	0	0.2655
Vehicle	4	Matched	11/11	13	2	0.3590
Vehicle	5	Matched	7/7	4	3	0.3662
Vehicle	6	Matched	2/2	1	0	0.2655
Vehicle	7	Random	11/11	23	4	0.3488
Vehicle	8	Random	7/7	9	0	0.3380
Vehicle	9	Random	2/2	10	0	0.2522
Vehicle	7	Trip	11/11	23	4	0.3488
Vehicle	8	Trip	7/7	9	0	0.3380
Vehicle	9	Trip	2/2	10	0	0.2522
Vehicle	7	Matched	11/11	23	4	0.3488
Vehicle	8	Matched	7/7	9	0	0.3380
Vehicle	9	Matched	2/2	10	0	0.2522
Scenario	1	Random	4/11	0	2	0.3902
Scenario	4	Random	6/11	2	5	0.3590
Scenario	7	Random	5/11	7	1	0.3488
Scenario	1	Trip	4/11	0	2	0.3902
Scenario	4	Trip	6/11	2	5	0.3590
Scenario	7	Trip	5/11	7	1	0.3488
Scenario	1	Matched	7/11	1	6	0.3902
Scenario	4	Matched	9/11	9	5	0.3590
Scenario	7	Matched	9/11	10	4	0.3488
Scenario	2	Random	2/7	4	0	0.3405
Scenario	5	Random	5/7	3	1	0.3662
Scenario	8	Random	6/7	2	4	0.3380
Scenario	2	Trip	2/7	4	0	0.3405
Scenario	5	Trip	5/7	3	1	0.3662
Scenario	8	Trip	6/7	2	4	0.3380
Scenario	2	Matched	7/7	8	1	0.3405
Scenario	5	Matched	7/7	9	0	0.3662
Scenario	8	Matched	7/7	9	0	0.3380
Scenario	3	Random	2/2	2	1	0.2919
Scenario	6	Random	2/2	0	0	0.2655
Scenario	9	Random	2/2	2	0	0.2522
Scenario	3	Trip	2/2	2	1	0.2919
Scenario	6	Trip	2/2	0	0	0.2655
Scenario	9	Trip	2/2	2	0	0.2522
Scenario	3	Matched	2/2	0	1	0.2919
Scenario	6	Matched	2/2	0	0	0.2655
Scenario	9	Matched	2/2	5	1	0.2522

Table S43: PVS pooled sensitivity with a 0.5-second matching tolerance.

Axis	Selector	Hits/intervals	Non-event	Redundant	Non-event/h
Vehicle	Random	59/60	66	19	22.36
Vehicle	Trip	59/60	66	19	22.36
Vehicle	Matched	59/60	66	19	22.36
Scenario	Random	34/60	21	15	7.11
Scenario	Trip	34/60	21	15	7.11
Scenario	Matched	52/60	50	19	16.94

The same predictions, annotation intervals and 2.9522 non-bump hours per axis are used. Total unmatched counts are unchanged; expanding event eligibility reclassifies some unmatched alarms as redundant event-associated notifications.

S17. Independent-source Karlsruhe validation

The original archive `Daten2018.zip` is 546,803,578 bytes; its official MD5 is `e9a4eb2d a043c1eb3dbf57134cf8796b`, with all archive member CRC checks passing. The retained 53 IMU workbooks contain 1,764,985 timestamped rows. The 13 additional event excerpts are excluded because some contain samples already present in the IMU tables. Header checks identify acceleration in C:E, gyroscope in F:H, GPS timestamp in S, coordinates in T:U, speed in V, material in AA and event code in AB. Six-axis features are recomputed from these measured channels. The source event labels used are 1 (good) and 3 (pothole), following Masino et al. (2018); other codes remain excluded.

GPS timestamps establish three usable date groups. Five workbooks without a recoverable GPS calendar are excluded before feature construction. Monotone sensor time supports within-file duration, while its differing epoch prevents a sensor-epoch calendar interpretation. The date/file mapping is retained with full arrays in the local audit. A file is a release container, and the date groups do not identify unique roads or new vehicles.

The frozen protocol is `KIT-PUBLIC-INDEPENDENT-SOURCE-20261103-v1`, SHA-256 `2782b0f43fd442a408f9acb8e6521cf7e64432abec45172a800c3983bb9d9105`. Schema, date/class counts and geographic overlap were inspected before this protocol; all model outcomes followed its freeze. The final date is excluded from candidate selection. Training windows are spatially purged using every raw coordinate in their span against eligible held-out anchors, followed by fixed common per-class inner training counts. Exact split and fit arrays are retained. The two date validation pools partition the same development windows as the random folds. Their full-window minimum separations are both 2,432.56 m. The final test has 917 windows and a retained-training separation of at least 922.43 m.

Independent checks confirm labels, sample features, full-window buffers, class budgets, candidate choice, all saved metrics, model reload predictions and selection-file precedence over final predictions. Twenty-four fit-folds implement the four fixed candidates across the two selectors and two outer refit targets.

The final engineering population comprises 12 evaluable pothole annotation intervals and 0.0578948 h of assessed good-labelled travel. Unknown and other event classes are outside the binary task, including their travel exposure. Alarm decisions use the 0.25-s decision grid; raw row differences vary with sensor cadence. A good-labelled raw row contributes its positive time increment capped at 0.1 s, with the final interval row using the median positive increment. Non-event alarms and redundant event-associated notifications are reported separately. No travelled-distance denominator or independently video-adjudicated impact time is available in this analysis. Both tolerance settings and all per-file counts are supplied below.

Table S44: Karlsruhe window populations before and after final-test spatial purging.

GPS date / pool	Role	Good	Pothole
13 May	Development	2,511	149
20 June	Development	1,613	36
5 July	Final test	835	82
Purged development	Full outer refit	4,123	166
Each actual inner fit	Shared budget	1,613	17

All dates are in 2016. Windows are one second long with 0.25-s decision spacing. Other labels and undated workbooks are excluded.

Table S45: Karlsruhe fixed-candidate estimates and final full-development scores.

Candidate	Random inner F1	Date inner F1	Final F1	Pothole recall
RF inertial	0.8320	0.5682	0.7978	0.4512
RF + context	0.9331	0.5433	0.6170	0.1585
HGB inertial	0.8554	0.5783	0.7855	0.4268
HGB + context	0.9598	0.5215	0.6782	0.2439

Candidate selection uses the corresponding inner score only. All final candidate outcomes are retained for audit; outer outcomes do not determine selector choices. Inertial denotes the 30 sensor statistics.

Table S46: Karlsruhe final-test per-file interval and notification counts.

Selector	Tolerance (s)	File ID	Hits / intervals	Non-event clusters	Redundant	Good h
Random	0.0	49	0/0	0	0	0.05789
Random	0.0	50	0/5	0	0	0.00000
Random	0.0	51	1/1	0	0	0.00000
Random	0.0	52	6/6	0	0	0.00000
Random	0.5	49	0/0	0	0	0.05789
Random	0.5	50	0/5	0	0	0.00000
Random	0.5	51	1/1	0	0	0.00000
Random	0.5	52	6/6	0	0	0.00000
Date	0.0	49	0/0	0	0	0.05789
Date	0.0	50	0/5	0	0	0.00000
Date	0.0	51	1/1	0	0	0.00000
Date	0.0	52	6/6	0	1	0.00000
Date	0.5	49	0/0	0	0	0.05789
Date	0.5	50	0/5	0	0	0.00000
Date	0.5	51	1/1	0	0	0.00000
Date	0.5	52	6/6	0	1	0.00000

Zero intervals are shown as 0/0, a count rather than a defined recall. File IDs are preserved archive-container identities, not independent roads. Assessed good exposure is concentrated in file 49.

S18. Supporting experimental comparisons

This section reports label populations, context-feature comparisons, the explicit bump/pothole task, count-matched PVS refits and Mexico/Puebla selections. Their protocols and domain-specific outcomes are given in Sections S2–S5 and S11–S15.

Table S47: RoadSens supervised populations under the two label policies.

Policy	Normal	Pothole	Bump	Total supervised rows
Primary	4,407	11,346	17,132	32,885
Blank-as-normal sensitivity	33,252	11,346	17,132	61,730

Table S48: RF feature ablations: five-fold mean macro-F1, seed 42. Sensitivity denotes the blank-as-normal policy.

Feature group	Primary random	Primary temporal	Primary recording	Sensitivity random	Sensitivity temporal	Sensitivity recording
GPS values	0.9985	0.8672	0.6984	0.9546	0.6180	0.3508
Weather	0.5465	0.5138	0.2958	0.4082	0.3586	0.2279
GIS elevation	0.6188	0.4354	0.3037	0.5533	0.4217	0.3888
Numeric minus context	0.9362	0.6739	0.2905	0.7792	0.4431	0.3362
IMU + GPS	0.9740	0.7953	0.7113	0.8398	0.5404	0.3929
IMU + weather	0.7814	0.6120	0.3059	0.5993	0.4155	0.3149
GPS availability	0.5233	0.5566	0.5431	0.2228	0.2310	0.2315
Numeric + availability	0.9926	0.8003	0.6797	0.9175	0.5683	0.3885

Table S49: Explicit bump/pothole classification: RF100 five-fold mean macro-F1.

Features	Random	Temporal	Recording
Full numeric	0.9948	0.7271	0.5189
Context	0.9991	0.8153	0.5559
Availability only	0.4143	0.4089	0.4029
Instantaneous IMU	0.6321	0.5107	0.5240

Table S50: Post hoc PVS common-budget diagnostic: three-domain means. The common-budget and historical-full-budget outer targets describe different refitted systems.

Axis	Selector	Inner F1	Outer F1	MAE
Common budget outer target				
Vehicle	Random-window	0.8581	0.8468	0.0140
Vehicle	Leave-one-trip	0.8402	0.8468	0.0217
Vehicle	Matched	0.8289	0.8526	0.0267
Scenario	Random-window	0.8699	0.7083	0.1627
Scenario	Leave-one-trip	0.8389	0.7141	0.1428
Scenario	Matched	0.7740	0.6951	0.1724
Historical full budget outer target				
Vehicle	Random-window	0.8581	0.8485	0.0400
Vehicle	Leave-one-trip	0.8402	0.8485	0.0477
Vehicle	Matched	0.8289	0.8522	0.0390
Scenario	Random-window	0.8699	0.7986	0.0941
Scenario	Leave-one-trip	0.8389	0.8064	0.0768
Scenario	Matched	0.7740	0.8168	0.1462

Table S51: Mexico/Puebla full-development refit outcomes by documented held-out route. Both selectors share the same selected outer model.

Held-out route	Inner selector	Chosen	Inner F1	Outer F1	Abs. error
Urban (1+2)	Random	RF	0.9922	0.9934	0.0012
Urban (1+2)	Whole-trip	RF	0.9922	0.9934	0.0012
Suburban (3+4)	Random	RF	0.9934	0.9960	0.0026
Suburban (3+4)	Whole-trip	RF	0.9950	0.9960	0.0010

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